

HUI PENG JOHN YAO

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EDUCATION

UNIVERSITY OF CALIFORNIA, LOS ANGELES

Los Angeles, CA

Bachelor of Science in Computer Science

Expected June 2028 | GPA: 3.6/4.0 | Dean's Scholarship

Coursework: Data Structures & Algorithms, Software Design & Construction, Computer Organization, Logic Design of Digital Systems

THOMAS JEFFERSON HIGH SCHOOL FOR SCIENCE AND TECHNOLOGY (TJHSST)

Alexandria, VA

June 2025 | GPA: 4.55 (Weighted) | SAT: 1580

Honors: National Merit Scholarship Semifinalist • Science Olympiad – 20+ regional, state, and national medals

Coursework: Artificial Intelligence, Mobile & Web Application Development, Linear Algebra, Probability & Statistics

PROFESSIONAL EXPERIENCE

DATAFORMA

York, PA

Software Engineering Intern | *Vue.js, Kotlin, SQL, AWS, Docker, Gradle*

June 2026 – Present

- Contributed to full-stack development for Dataforma, a commercial field service management platform.
- Developed Vue.js front-end features to improve usability and customer workflows.
- Implemented SQL-based querying solutions in Kotlin to support additional user-facing data formats.
- Created Docker images to enable consistent local backend development and testing.
- Built and optimized Gradle pipelines integrating multi-language services into a streamlined build process.
- Leveraged AWS S3 and EC2 services to support application storage and deployment.

WOOLLY

Los Angeles, CA

Full-Stack Software Engineer | *React, Express, Monaco Editor, Python, SQL, AWS*

April 2026 – Present

- Developed full-stack features for an AI-powered recruiting platform that evaluates real-world software engineering skills beyond traditional coding interviews.
- Engineered a browser-based AI coding environment using Monaco Editor that retrieves project context, integrates Claude, and automatically evaluates prompts, generated code, token usage, and project completion time to assess AI-assisted coding performance.
- Created six interactive assessment modules using React and Express to evaluate candidates' data analysis, presentation, communication, and writing skills.
- Processed and transformed assessment data using Python and SQL to support evaluation workflows, and implemented Python-based unit tests to ensure reliable and standardized deployment.
- Proposed software architectures, presented designs to the engineering team, and collaborated on design decisions.

FITICA

Los Angeles, CA

Founder & Lead Software Engineer | *React Native, Node.js, AWS, Flowise, Expo, OAuth*

June 2025 – Present

- Founded and led a five-member engineering team developing an AI-powered fitness platform that delivers personalized fitness coaching through a mobile app.
- Designed an AI architecture using Flowise agents, long-term user memory, and API integrations to generate personalized workout plans that adapt to user performance.
- Developed an AI-powered nutrition analysis pipeline that processes food images through Flowise to generate calorie estimates and personalized nutrition recommendations.
- Engineered a cross-platform mobile application using React Native and Expo.
- Built and deployed scalable backend services using Node.js and AWS (Lambda, DynamoDB, API Gateway), implementing OAuth-based authentication and role-based access control to secure user data.

UNIBOUND CONSULTING

Fairfax, VA

Co-Founder and Lead Consultant | *React, Tailwind, Node.js, Express*

May 2025 – June 2026

- Co-founded a college consulting startup that generated approximately \$15,000 in revenue.
 - Led front-end development of a client-facing website that showcases consulting services and enables students to schedule advising sessions.
 - Provided one-on-one college consulting services, helping students develop competitive college application strategies.
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PROJECTS

A-Eye – Winner, Best Use of ElevenLabs, LA Hacks 2026 (April 2026) | *React, YOLO, Cloudflare, ElevenLabs*

- Led a four-person team to develop prototype smart glasses for visually impaired users.
- Applied YOLO object detection, multi-agent decision making, and ElevenLabs text-to-speech to identify obstacles, detect points of interest, and provide real-time audio guidance.

Infinite Delve (June 2025 – May 2026) | React, Node.js, AWS, PostgreSQL, Flowise

- Developed an AI-powered narrative game where generative AI dynamically creates crafting outcomes, item progression, and narrates player interactions with the game world.
- Built a full-stack cloud application using React, Node.js, PostgreSQL, and AWS to support persistent inventory, player progression, and game state.

Alexa-Powered Journal Assistant (Dec. 2024 – Feb. 2025) | Python, AWS Lambda, Google Drive API, DynamoDB, Alexa Skills Kit, OAuth 2.0

- Built an AWS Lambda application that securely retrieves Google Docs journal entries using OAuth 2.0 and stores structured data in DynamoDB.
- Developed and published an Alexa Skill integrating Google Drive, AWS Lambda, and DynamoDB to enable voice-based access to journal entries and to-do lists.

Arcane Chess (Sept. 2023 – May 2024) | Godot, GDScript

- Developed a desktop strategy game using Godot that combines traditional chess with card-based mechanics.

Published Research (Oct. 2023 – May 2024) | R, Python, Statistical Analysis

- Designed and distributed a psychological survey using the Brief-COPE inventory, collecting 139 responses.
- Published findings in the Whitman Journal of Psychology after performing statistical analyses in R and Python using ANOVA, paired t-tests, and chi-squared tests.

SKILLS & CERTIFICATIONS

Certifications: AWS Certified Cloud Practitioner ▪ AI Agents: Automation & Business with LangChain & LLM Apps (Udemy) ▪ The Complete Full-Stack Web Development Bootcamp (Udemy)

Programming Languages: Python, Java, JavaScript, Kotlin, C++, R, GDScript, HTML/CSS

Frameworks & Libraries: React, React Native, Vue.js, Node.js, Express, Tailwind, Bootstrap, PyTorch, Django

Cloud & Databases: AWS (Lambda, EC2, S3, DynamoDB, IAM, API Gateway, CloudFront, Route 53, Certificate Manager), PostgreSQL, SQL

AI & Data Analysis: NumPy, pandas, R, Statistical Testing (ANOVA, t-tests, chi-square), Data Visualization, Computer Vision, Flowise

Tools & Platforms: Git, Docker, Gradle, REST APIs, OAuth 2.0, Linux, Alexa Skills Kit, Google Drive API, Cursor AI, Slack, Monaco Editor, ElevenLabs, Claude, Copilot

LEADERSHIP

Development Team Officer (Nov. 2025 – Present) | GLITCH @ UCLA

- Developed an authentication portal to streamline participant access for a university hackathon.
- Designed and implemented OAuth-based authentication flows and backend services using AWS API Gateway.
- Collaborated with organizers to define technical requirements and prioritize platform features.

Woodwind Captain (Sep 2021 – May 2025) | TJHSST Marching Band

- Led and mentored 40+ musicians, developing lesson plans and onboarding materials for new members.
- Organized sectionals and team-building activities, including custom card games, to strengthen collaboration and section cohesion.

Science Olympiad Team Lead (Sep 2021 – May 2024)

- Led and competed in engineering and research events, including Experimental Design and Thermodynamics.
- Earned 20+ medals across regional, state, and national competitions through collaborative problem-solving.